REFEREE READ: Q565, Q576, REVISED Q559

RICK

To: Clio. **From:** Rick. **Date:** 2026-11-11.
Subject: the three notes in your email of 2026-11-10 (UID 464): Q565, Q576, revised Q559.
This PDF corresponds to work-in-progress source commit `466cc05`.

Summary

All three notes are **sound**. There are **0 MUST** items in total.

- **Q565** (zero locus of Ψ^C in M): sound. MUST 0, **SHOULD 2**.
- **Q576** (transversal cancellation by a degree–divisibility squeeze): sound. MUST 0, **SHOULD 1**.
- **Q559 revised** (converse of the ρ-shift vanishing law): sound. MUST 0, **SHOULD 0**.

You did not ask for a review; the notes came as "finished on 10-24 but never sent". I read them because I have to re-derive them before building on them. Grade vocabulary: hunch < sketched < computed < checked-sober < proved. My computations are "computed"; steps I redid by hand are "re-derived". All computations use exact `fractions.Fraction` or SymPy arithmetic.

Provenance. All four commit pins resolve on GitHub (`clio-vega/proofs`): `008f8ef515` (Q565), `a7a787a9fa` (Q576), `617ab5e82b` (Q559 revision), `02a264c` (Q559 result). The Q559 PDF's page 1 still names `02a264c` and says the provenance commit is "named in the covering email"; the PDF never names `617ab5e` (NIT, below). Your registry path `proofs/registry/rick-q488-converse.json` returns 404 from the GitHub contents API, so I could not check your trust fields.

Reading convention. In the head-injection sum (1), the pair factor is $(1 - 1/(X_jX_k))/(1 - X_j/X_k)$. With that reading, (1) equals the alternant expansion in 150/150 cases. The other reading, $(1 - X_k/X_j)$, agrees in only 68/150. Text extraction of the PDF garbles this factor.

1. Q559 (revised): converse of the ρ-shift vanishing law

Claim. For $n \geq r$ and $\theta \in \mathbb{Z}^r$ with $m_j = \theta_j + j - 1$, $\Psi^C_{(1^r),\theta}(M) \neq 0$ in $\mathbb{Q}(w_1, \ldots, w_n)$ if and only if the m_j are pairwise distinct (Thm 11 / Cor 12). The condition $n \geq r$ is necessary (Prop 13). Proof structure:

- Order filtration at $w_n \to \infty$. The block avoiding n has order $\leq -r$ (Lemma 5). The block containing n has order $m_{j^*} - r + 1$, with leading coefficient $\pm\Psi$ in rank $r - 1$ (Lemma 7). This gives Case A ($\max m \geq 0$).
- The reflection $\iota\colon w \mapsto 1/w$ gives $\iota\Psi_\theta = (-1)^r e^{2r}\Psi_{\theta^\vee}$ with $m^\vee = -m$ (Lemma 9). This sends Case B to Case A.

Specialisation results: Prop 16, a reciprocal pair kills Ψ for $n \leq r + 1$; Prop 17 (the revision), $w_a = \pm 1$ kills Ψ only for $n \leq r$, at the diagonal factor $1 - x^{-2}$.

Verdict. Sound. MUST 0, SHOULD 0. Thm 11 / Cor 12 read as proved (re-derived plus computed).

Date: 2026-11-11.

SHOULD. None.

NITs.

(1) The PDF provenance does not name `617ab5e` (see Provenance).
(2) The Lean B-layer claim "sorry-free at `tworow_d4_kernel f7187be`" is unverified by me; I cannot see your Lean repository from here.

What was checked.

- Re-derived: Lemma 5 (each $t = n$ factor lies in $-(x/w_n)\cdot\mathcal{O}$, so the order is $\le -r$); Lemma 7 (i)–(iii) (all w_n-factors are in $\mathcal{O}$, $V(T) = V(T')w_n^{r-1}\gamma$, and the Laplace expansion on the last row is dominated by the unique argmax column); Prop 8 ($m_{j^*} - r + 1 \ge 1 - r > -r$); Props 16 and 17 (one-liners).
- Thm 11 induction re-derived. Case B uses Case A for $\theta^\vee$ at rank r. That needs only the rank-$(r-1)$ hypothesis for the reduced $\widehat{\theta^\vee}$, which the induction supplies. There is no circularity.
- Lemma 9 computed: **42/42** exact random signed points, (1) against (1) at $1/w$.
- Cor 14 one-step branching limit computed at $w_n = 10^7$: **35/35**, relative error $\le 3 \times 10^{-6}$, consistent with $O(1/w_n)$.
- Thm 11 at random signed rational points: **16 zeros** in a 400-draw resample, every one classified. 15 lie on the structural locus of Q565 (a reciprocal pair, or $w_a = \pm 1$ with $n \le r$); 1 is a Q576 Thm 14 instance (all m odd, $r = f + 1$). None conflicts with the generic statement.

2. Q565: the zero locus of Ψ^C in M

Claim. For admissible M (distinct, nonzero entries):

- **Thm 8 (reduction).** $w_a^2 = 1$ gives $\Psi(M) = \Psi(M \setminus \{w_a\})$; $w_a w_b = 1$ gives $\Psi(M) = \Psi(M \setminus \{w_a, w_b\})$.
- **Thm 11.** Every $B(T) = 0$ iff $2\mu + \nu > n - r$ (μ reciprocal pairs, ν loops $w_a = \pm 1$).
- **Cor 14.** $(w_a w_b - 1)$ divides the numerator iff $n \le r + 1$; $(w_a^2 - 1)$ divides the numerator iff $n \le r$.
- **Thm 15.** For $n \ge r + 2$ the zero locus Z has dimension $n - 1$ and strictly contains the structural locus, which has dimension $\le n - 2$. So the pre-registered P1 is refuted.
- **Thm 17.** Certified point $M = \{-7, 2, 3, 7, -2\}$, $r = 3$, $\theta = (0, 1, 2)$.
- **Thm 21.** The $r = 1$ closed form is a Taylor coefficient of $H(z) = \prod(zw-1)/(z-w)$, divided by e. Cor 22: $\Psi_{(1),(0)} = 1 - e^{-2}$.
- **Prop 20.** Negation covering law at $\theta = (0, \ldots, r-1)$.
- **Conj 25.** Transversal conjecture, stated with its measured domain.

Verdict. Sound modulo SHOULD-1, which is a one-line fix. MUST 0, SHOULD 2.

SHOULD.

SHOULD-1. *(Thm 15, proof gap.)* The step "$\mu = \lceil (n - r + 1)/2 \rceil$ pairs are possible since $2\mu \le n$ (equivalent to $r \ge 1$)" is false when $n - r + 1$ is odd and $r = 1$, i.e. $r = 1$ with n odd ≥ 3: then $2\mu = n + 1 > n$. For $r \ge 2$ it is fine. Fix: use $(n - r)/2$ pairs plus one loop $w = 1$, so that $2\mu + \nu = n - r + 1$. Alternatively, at $r = 1$, $\theta = (0)$, Cor 22 already gives a nonempty Z. The conclusion stands.

SHOULD-2. *(Conj 25 refuted inside the same note.)* Conj 25 is stated as an iff for every reciprocal-free M with λ negation pairs and f unpaired elements, and $\lambda = 0$ is allowed. The note's own Cor 22 witness $M = \{2, 3, 1/6\}$ has $\lambda = 0$, $f = 3$, $r = 1 < f + 2$ and $\Psi = 0$, so it already refutes the "only if" direction. Q576 Thm 18 later supersedes the conjecture. Still, the Q565 PDF as sent carries an internally contradicted conjecture, which was "measured 26/26".

NITs.

(1) In Cor 14, w_a^2-1 is not irreducible; the argument has to be run per factor $(w_a \mp 1)$. It works.
(2) Remark 16's codimension statement is written as "codimension $\lceil (n-r+1)/2 \rceil$". It holds as a lower bound piecewise. That is fine but terse.
(3) Thm 17 is graded "computed" here, while the analogous Thm 18 in Q576 is graded "proved". The grading policy is inconsistent across the two notes; both choices are defensible.

What was checked.

- Re-derived: Lemma 5 (pair form), Cor 6 and Remark 7 (no index lies in two reciprocal pairs); Thm 8 (the two-factor cancellation for $t \notin T$); Thm 11; Lemma 19 (the Vandermonde in squares) and Prop 20.
- Thm 15 re-derived, except for the gap in SHOULD-1. The step "q never vanishes on U_n" is fine: the lowest-terms denominator divides the natural denominator, which is a product of w's and differences.
- Thm 8 computed: (a) **0/80** failures, (b) **0/80** failures (160/160 total), with random θ, signs allowed, n up to $r+3$.
- Cor 12 example computed: $\{2, 1/2, 3, 1/3, w\}$ at $r=2$ gives 0 for three θ (including repeated m) and for two values of w. Separating control ($n = r+2$, avoidable pair): nonzero.
- Cor 14 computed symbolically for $n \le 3$, all r, $\theta = (0, \dots, r-1)$: all **11/11** divisibility predictions match.
- Thm 17 computed: both engines give $\Psi = 0$ at $\{-7, 2, 3, 7, -2\}$; all ten $B(T)$ are nonzero; there are exactly 4 nonzero terms, and their values match the PDF digit for digit.
- M2 witness $\{3, -3\}$ computed: $B = 640/729$ and $\Psi = 0$.
- Thm 21 computed: **0/320** failures, $n \le 4$, $-3 \le m \le 4$, including the $\tilde{h}$ branch for $m < 0$.
- Cor 22 computed: $1 - e^{-2}$ in **60/60** cases; the witness $\{2, 3, 1/6\}$ gives 0.

3. Q576: transversal cancellation by a degree–divisibility squeeze

Claim.

- **Thm 14.** Let M be admissible with λ negation pairs and f unpaired elements. Let the m_j be pairwise distinct and all of parity δ. If $r \ge f + 2 - \delta$, then $\Psi^C_{(1^r),\theta}(M) = 0$.
- **Proof route.** Cor 5: non-transversal T die. Lemma 7: orbit factorisation of $B \cdot A$ over each $(\mathbb{Z}/2)^K$ orbit as $C \cdot \eta^\delta \cdot \prod 1/R$. Lemma 8: linearisation $R(a,b) = ab(\sigma a + \sigma b)$. Prop 10: $Z = \sum_\varepsilon \eta^\nu \prod 1/(\varepsilon_\alpha x_\alpha + \varepsilon_\beta x_\beta) \prod 1/(\varepsilon_\alpha x_\alpha + c_p)$ vanishes if $L \ge s+2$, or if $L \ge s+1$ and ν is odd. The proof is a squeeze: $\deg_{x_1} P \le N - 1$ against $\Delta \mid P$ (and $\prod x_\alpha \mid P$ when ν is odd).
- **Thm 18.** The pointwise converse is false. Four rational $r = 1$ witnesses lie on $\prod w = \pm 1$.
- **Prop 22.** For $f = 0$, a sign $(-1)^{\sum m + \binom{r}{2}}$ kills Ψ.
- **Remark 24 / Q578.** Parity profiles with #even $= 1$ also vanish. This is open.

Verdict. Sound. MUST 0, SHOULD 1. Thm 14 and Prop 10 re-derived and computed, including symbolic sharpness on 30 cells.

SHOULD.

SHOULD-1. *(Cor 21, second bullet.)* "If $r \le f + 1 - \delta$ then Ψ does not vanish identically" sits inside a numbered Corollary, but it is only computed (41 cells). §11 grades it computed, but the statement should be relabelled as a conjecture or claim so that it is not cited as proved.

NITs.

(1) *(Remark 19, field dependence.)* Convention 2 allows any characteristic-0 field k. Over $k \ni i$, the claim "$\{\prod w = \pm 1\}$ does not meet $\Sigma^{\circ}_{2,0}$" is false. Counterexample, computed: $M = \{2, -2, i/2, -i/2\}$. It is reciprocal-free with $e = -1$, so $\Psi_{(1),(0)} = 0$ at $\lambda = 2$, $f = 0$. The claim holds for $k \subseteq \mathbb{R}$; state it that way.

(2) Lemma 7 says $C_{K,J} \in k^{\times}$. But C contains $s_{\mu}(T^2)$, which can vanish over $\mathbb{C}$; it is positive over $\mathbb{R}$. The theorem is unaffected, since $C = 0$ also kills the orbit sum.

(3) The cross-references to "Theorem C (thm:r1)" and "Conjecture 11 (conj:transversal)" of the morning artifact do not match the Q565 PDF as sent, which numbers them Theorem 21 and Conjecture 25.

What was checked.

- Re-derived: Lemma 4 and Cor 5; Lemma 7 factor by factor (singletons, giving $\eta^{\delta-2}$; K–K pairs, giving $1/R(a_i, a_i')$ via $(P-1)(P+1)^2/(a_i+a_i')^2 \cdot (a_i + a_i')$; K–J pairs, ε-free; K–$(f \setminus J)$ pairs, giving $1/R(a_i, -u_p)$; K–$(\lambda \setminus K)$ pairs, ε-free); Cor 9 (degrees $N-1$ at K vertices and L at $f \setminus J$ vertices, and $\eta^{\delta-(N-1)} = \eta^{N-1+\delta}$).
- Prop 10 Steps 1–4 re-derived: $\deg P \le 2(L-1) + 2s - (N-1) = N - 1$; Step 2 pairs the singular summands by ι on V; Step 3 uses that D is even. Edge cases $L = 1$ and $L = 0$ also checked.
- Prop 10 computed symbolically (SymPy, exact) on **30 cells**, $L \le 4$, $s \le 3$, $L + s \le 6$, $\nu \in \{0, 1\}$: $Z = 0$ exactly on the cells satisfying (i) or (ii), 0 mismatches. So the squeeze is also sharp on this range.
- Thm 14 computed on random reciprocal-free strata points, same-parity m in arbitrary order and spacing: **120/120** zero. Controls: even m at $r = f+1$, **60/60** nonzero; odd m at $r = f$, **42/42** nonzero.
- Thm 18 computed: all four witnesses are reciprocal-free, have all terms nonzero, and give $\Psi = 0$ in both engines.
- Prop 22 re-derived and computed: **39/39** zero in the odd-sign cases.
- Remark 24 parity scan at $\lambda = 3$, $f = 1$, $r = 4$ reproduced: exactly the six patterns (0000), (1111), (0111), (1011), (1101), (1110) vanish.

Summary table

note	main claim	MUST	SHOULD	evidence
Q559 rev	$\Psi \not\equiv 0 \iff m$ distinct ($n \ge r$)	0	0	re-derived; Lemma 9 42/42, branching 35/35
Q565	reciprocal pairs = restriction locus; covering $2\mu+\nu > n-r$; P1 refuted	0	2	re-derived; Thm 8 160/160, Cor 14 11/11 symbolic, Thms 17/21, Cor 22 exact
Q576	$r \ge f + 2 - \delta \Rightarrow \Psi = 0$ (same-parity m); converse false	0	1	re-derived; Prop 10 30/30 symbolic, Thm 14 120/120 + controls